1 Differentiation

1. if $y = (x + \sqrt{x^2+1})^2$, show that 7. if $y = (x + \sqrt{x^2+1})^2$, show that $\frac{dy}{dx} = \frac{2y}{\sqrt{x^2+1}}$

2 Applications of differentiation

1. Find the smallest angle between the tangent to the graph of f at $x = 1$ and the x axis
2. Find the acute angle between the line $y = -3x + 13$ and the tangent to f at $x = \frac{1}{2}$
3. City A is located at the point (0, 0) and city B is located at the point (10, 0). Let

$$f(x) = \frac{p}{x+1} + \frac{4}{11-x}, x \in [0, 10]$$

   represent the pollution level between the two cities.

   (a) Find value(s) of p such that city B is the most polluted point between the two cities.

   (b) Find value(s) of p such that city B is the least polluted point between the two cities.

4. The concentration of insects per square metre in a gorge, C is a continuous function of time t, given by $C(t) = \begin{cases} 1000\left(\cos\left(\frac{\pi(t-8)}{2}\right)+2\right)^2 - 1000 & 8 \le t \le 16 \\ m & 0 \le t < 8 \text{ or } 16 < t \le 24 \end{cases}$

   Where t is the number of hours after midnight and $m \in \mathbb{R}$

   (a) What is the value of m?

   (b) Sketch the graph of C for $t \in [0, 24]$

   (c) What is the minimum concentration of insects and at what value(s) of t does that occur?

   (d) The insect infestation is considered deadly if the concentration is more than 1250 insects per square metre. At what time after midnight does the concentration of insects first stop being deadly?

   (e) During a 24 hour periiod, what is the total length of time for which the insect infestation is not deadly?

   (f) Due to the uneven surface of the gorge, the time, T minutes, it takes to run x km is given by $T = p(q^x - 1)$ where p and q are constants.

      i. If it takes 5 minutes to run the first kilometre and 12.5 mins to run the first two kilometres, find the values of p and q

      ii. Find the length of time it takes to run 4km into the gorge to retrieve a buried treasure

      iii. It takes 19 minutes to dig up the treasure and you are able to run back out of the gorge in half the time it takes to reach the treasure. Show that it is possible to recover the treasure successfully and state how much time there will be to spare.

5. The light intensity level L, in a greenhouse varies over the course of a day according the function below.

$$L(t) = \begin{cases} -\frac{1}{18}t^2 + \frac{4}{3}t - 6 & , \quad 6 \le t < 7 \cup 17 \le t < 18 \\ \frac{1}{4}\sin\left(\frac{3\pi t}{5}\right) + H & , \quad 7 \le t < 17 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

Where t is the number of hours after midnight and $H \in \mathbb{R}^+$.

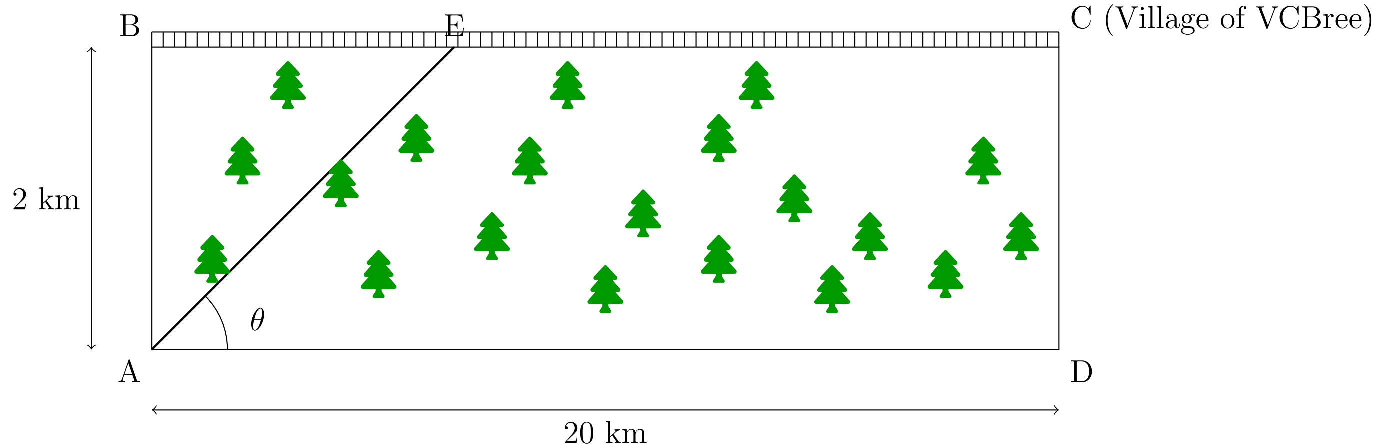
(a) Assuming $L(t)$ is a continuous function, find the value of H

(b) Sketch the graph of L for $t \in [0, 24]$

(c) Find the maximum light intensity level and at what value(s) of t does that occur?

(d) For what time interval(s) is the light intensity in the top 10% of the intensities throughout the day?

6. The hobbits, on their way to the village of VCBree find themselves point A on the south side of a rectangular forest that is 2km wide, and need to get to point to the village VCBree located at point C, 20km to the east and on the other side of the forest. Point B is directly to the north of point A and on the edge of the forest as shown. There is a track running along the far side BC of the forest. They need to get from A to C. They can travel at a speed of 24km/hr on the track but only at a speed of 12km/hr through the forest. Note: The Diagram is not to scale.



(a) Calculate the time taken to travel directly from A through the forest to B and then from B to C using the track.

(b) Find the distance the hobbits travel inside the forest from A to E as a function of θ

(c) Explain why the possible values of θ, correct to the nearest degree, are $\theta \in [6^o, 90^o]$

(d) Find the time taken to travel from A to C through the forest, as a function of θ

(e) Find the value of θ that minimises the time taken to travel from A to C through the forest and calculate the minimum time taken.